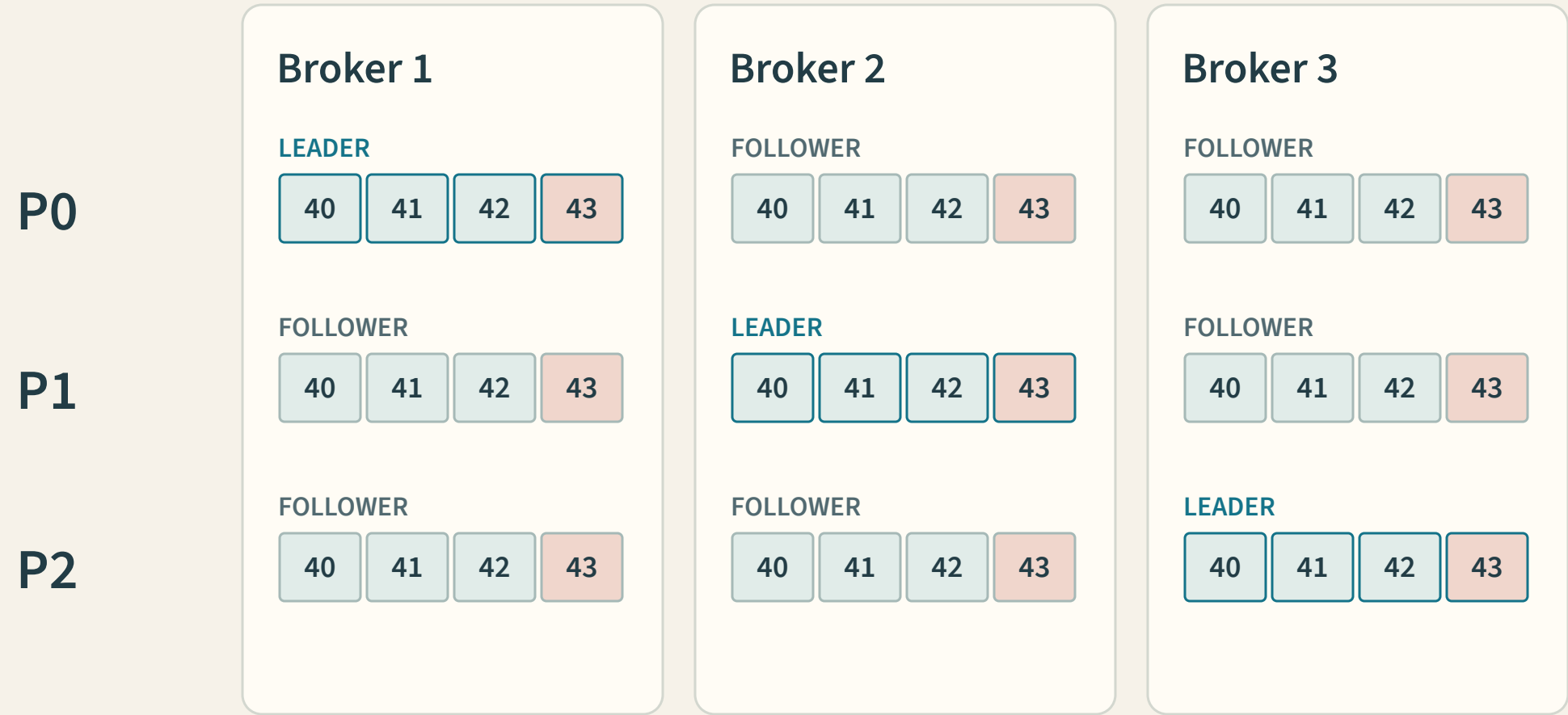


Apache Kafka: four views of a durable event log

Partition order, replica copies, independent consumers and metadata consensus are separate parts of the same system.

01 One topic, three ordered logs

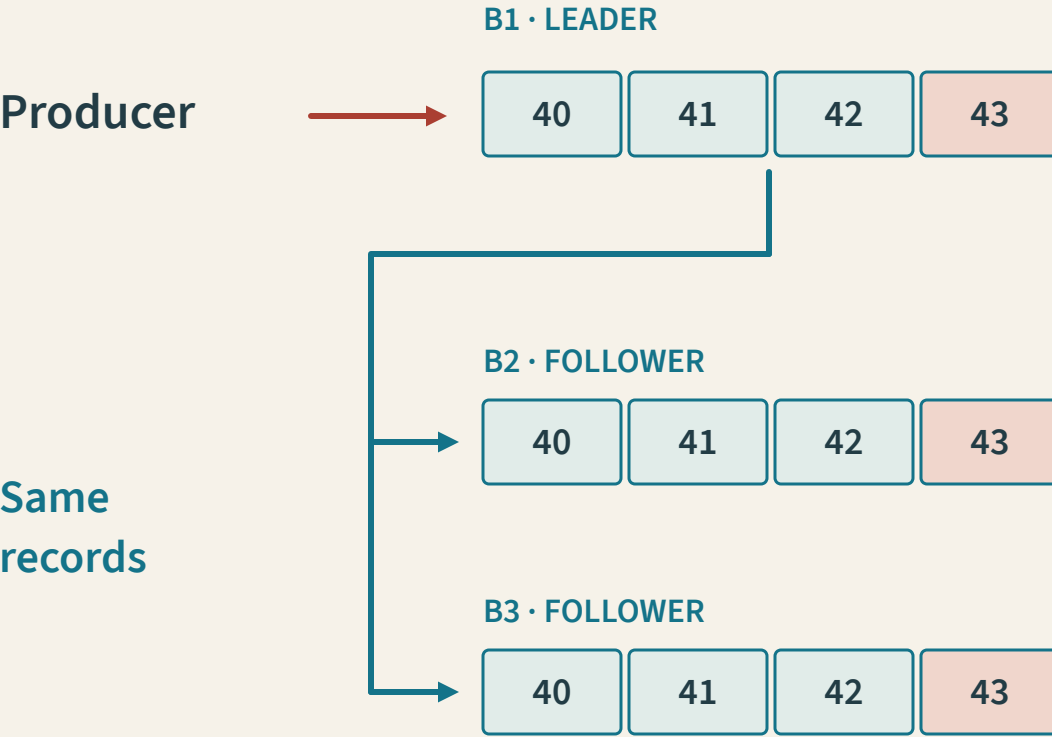
Rows are partitions. Columns are brokers. Every partition has three replicas.



Offsets order records within each partition. There is no single topic-wide order.

02 Zoom into partition P0

Arrows show record transfer; followers fetch from the leader.

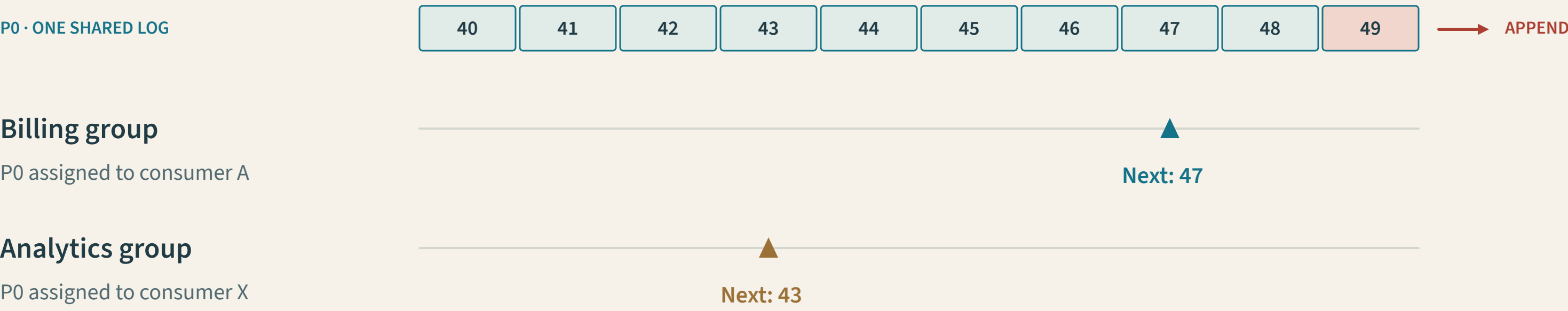


acks=all · min.insync.replicas=2

All current ISR members must replicate the write; fewer than two means failure.

03 Consumers move; retained records stay

Two independent standard consumer groups read the same topic, each with its own progress.



One assigned consumer per partition within each group. Replay is possible while records remain retained.

04 Cluster metadata has a different consensus path

A dedicated KRaft controller quorum tracks brokers, topics and partition leadership.



A majority of three voters tolerates one controller failure.

Two durability mechanisms

KRaft commits metadata through a voter majority. Topic records use partition replication and the ISR contract above.

Consumer offsets live in a broker-hosted internal topic, `__consumer_offsets`.